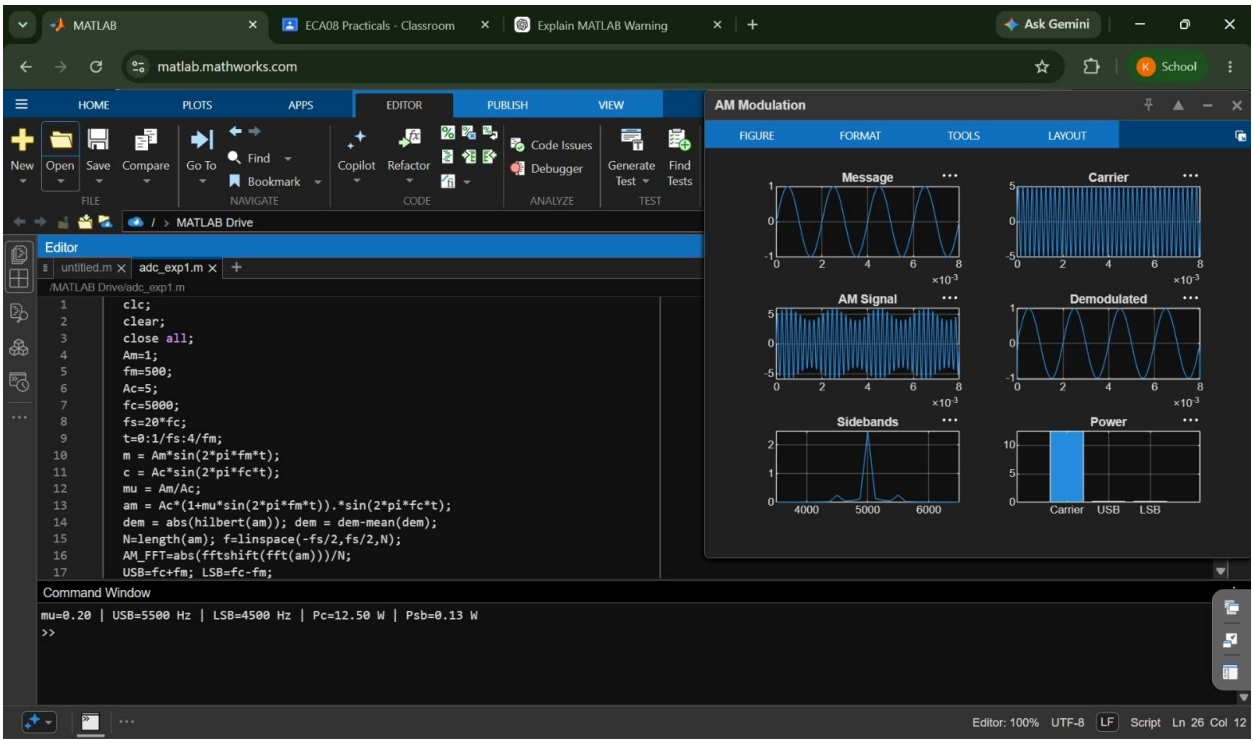
Experiment -2

Code:

```
clc;
clear;
close all;
Am=1;
fm=500;
Ac=5;
fc=5000;
fs=20*fc;
t=0:1/fs:4/fm;
m = Am*sin(2*pi*fm*t);
c = Ac*sin(2*pi*fc*t);
mu = Am/Ac;
am = Ac*(1+mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);
dem = abs(hilbert(am)); dem = dem-mean(dem);
N=length(am); f=linspace(-fs/2,fs/2,N);
AM_FFT=abs(fftshift(fft(am)))/N;
 USB=fc+fm; LSB=fc-fm;
Pc=Ac^2/2; Psb=(mu^2*Pc)/4;
figure('Name','AM Modulation','NumberTitle','off');
subplot(3,2,1), plot(t,m), title('Message'), grid on
subplot(3,2,2), plot(t,c), title('Carrier'), grid on
subplot(3,2,3), plot(t,am), title('AM Signal'), grid on
subplot(3,2,4), plot(t,dem), title('Demodulated'), grid on
subplot(3,2,5), plot(f,AM_FFT), xlim([fc-3*fm fc+3*fm]), title('Sidebands'), grid on
subplot(3,2,6), bar([Pc Psb Psb]), set(gca,'XTickLabel',{'Carrier','USB','LSB'}),
title('Power'), grid on
fprintf('μ=%.2f | USB=%d Hz | LSB=%d Hz | Pc=%.2f W | Psb=%.3f
W\n',mu,USB,LSB,Pc,Psb)
```


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ECA08 Practicals - Classroom
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Debugger
Generate Test
Find Tests
FILE
NAVIGATE
CODE
ANALYZE
TEST
MATLAB Drive
Editor
untitled.m
adc_exp1.m
/MATLAB Drive/adc_exp1.m
clc;
clear;
close all;
Am=1;
fm=500;
Ac=5;
fc=5000;
fs=20*fc;
t=0:1/fs:4/fm;
m = Am*sin(2*pi*fm*t);
c = Ac*sin(2*pi*fc*t);
mu = Am/Ac;
am = Ac*(1+mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);
dem = abs(hilbert(am)); dem = dem-mean(dem);
N=length(am); f=linspace(-fs/2,fs/2,N);
AM_FFT=abs(fftshift(fft(am)))/N;
USB=fc+fm; LSB=fc-fm;
Command Window
mu=0.20 | USB=5500 Hz | LSB=4500 Hz | Pc=12.50 W | Psb=0.13 W
>>
AM Modulation
FIGURE
FORMAT
TOOLS
LAYOUT
Message
Carrier
AM Signal
Demodulated
Sidebands
Power
Carrier
USB
LSB
Editor: 100%
UTF-8
LF
Script
Ln 26 Col 12